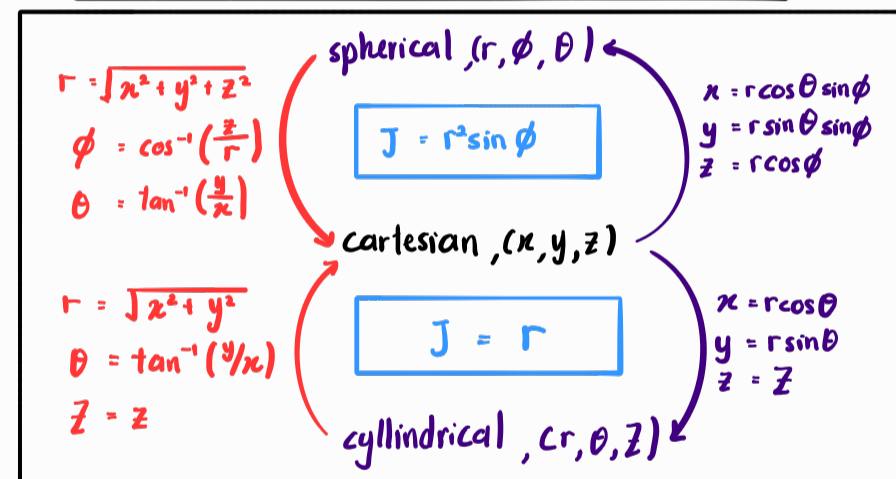


$\frac{d}{dx} [\log_a x] = \frac{1}{x \ln a}$	$\frac{d}{dx} [\cos x] = -\sin x$	$\frac{d}{dx} [\cosh x] = \sinh x$
$\frac{d}{dx} [\ln(ax+b)] = \frac{a}{ax+b}$	$\frac{d}{dx} [\sin x] = \cos x$	$\frac{d}{dx} [\sinh x] = \cosh x$
$\frac{d}{dx} [a^x] = a^x \ln a$	$\frac{d}{dx} [\tan x] = \sec^2 x$	$\frac{d}{dx} [\tanh x] = \operatorname{sech}^2 x$
$\frac{d}{dx} [\ln(ax+b)] = \frac{a}{ax+b}$	$\frac{d}{dx} [\sec x] = \sec x \tan x$	$\frac{d}{dx} [\operatorname{sech} x] = -\operatorname{sech} x \tanh x$
$\frac{d}{dx} [\cosh^{-1} x] = \frac{1}{\sqrt{x^2-1}}$	$\frac{d}{dx} [\operatorname{cosec} x] = -\operatorname{cosec} x \cot x$	$\frac{d}{dx} [\operatorname{cosech} x] = -\operatorname{cosech} x \coth x$
$\frac{d}{dx} [\sinh^{-1} x] = \frac{1}{\sqrt{x^2+1}}$	$\frac{d}{dx} [\cot x] = -\operatorname{cosec}^2 x$	$\frac{d}{dx} [\coth x] = -\operatorname{cosech}^2 x$
$\frac{d}{dx} [\tanh^{-1} x] = \frac{1}{1-x^2}$		

$\int f'(x) [f(x)]^n dx = \frac{[f(x)]^{n+1}}{n+1} + C$	$\int \sin(ax+b) dx = -\frac{1}{a} \cos(ax+b) + C$
$\int \frac{f'(x)}{f(x)} dx = \ln f(x) + C$	$\int \cos(ax+b) dx = \frac{1}{a} \sin(ax+b) + C$
$\int f'(x) e^{f(x)} dx = e^{f(x)} + C$	$\int \sec^2(ax+b) dx = \frac{1}{a} \tan(ax+b) + C$
$\int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + C$	$\int f'(x) \sin[f(x)] dx = -\cos f(x) + C$
$\int \frac{1}{ax+b} dx = \frac{1}{a} \ln ax+b + C$	$\int f'(x) \cos[f(x)] dx = \sin f(x) + C$
$\int e^{ax+b} dx = \frac{1}{a} e^{ax+b} + C$	$\int f'(x) \sec^2[f(x)] dx = \tan f(x) + C$

θ in degrees	0°	30°	45°	60°	90°	180°	270°	360°
θ in radians	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	0	-1	0
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	-1	0	1
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	undefined	0	undefined	0

$$\frac{dT(x,y,z)}{dt} = \frac{\partial T}{\partial x} \left(\frac{\partial x}{\partial t} \right) + \frac{\partial T}{\partial y} \left(\frac{\partial y}{\partial t} \right) + \frac{\partial T}{\partial z} \left(\frac{\partial z}{\partial t} \right)$$



$$m = \iiint_V \rho(x,y,z) dx dy dz$$

$$\bar{x} = \frac{1}{m} \iiint_V x \cdot \rho(x,y,z) dx dy dz$$

$$\bar{y} = \frac{1}{m} \iiint_V y \cdot \rho(x,y,z) dx dy dz$$

$$I_x = \iiint_V [(y^2 + z^2) \cdot \rho(x,y,z)] dx dy dz$$

$$I_y = \iiint_V [(x^2 + z^2) \cdot \rho(x,y,z)] dx dy dz$$

$$I_z = \iiint_V [(x^2 + y^2) \cdot \rho(x,y,z)] dx dy dz$$

$$\sin(A+B) = \sin A \cos B + \cos A \sin B$$

$$\cos(A+B) = \cos A \cos B - \sin A \sin B$$

$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\sin(2A) = 2 \sin A \cos A$$

$$\cos(2A) = \cos^2 A - \sin^2 A$$

$$= 1 - 2 \sin^2 A$$

$$= 2 \cos^2 A - 1$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$$

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

2D Laplace Egn

$$\nabla^2 u = 0$$

$$\text{Cartesian: } \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$$\text{Polar: } \frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0$$

Field Line, $\vec{r}(t) \Rightarrow \vec{F}(\vec{r}(t)) = \frac{d}{dt}(\vec{r}(t))$

$\nabla f(x, y, z) = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle$

$\nabla f(r, \theta, \phi) = \left\langle \frac{\partial f}{\partial r}, \frac{1}{r} \frac{\partial f}{\partial \theta}, \frac{1}{r \sin \theta} \frac{\partial f}{\partial \phi} \right\rangle$

$\nabla f(r, \theta, z) = \left\langle \frac{\partial f}{\partial r}, \frac{1}{r} \frac{\partial f}{\partial \theta}, \frac{\partial f}{\partial z} \right\rangle$

Divergence: $\nabla \cdot \vec{F} = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) \cdot \begin{pmatrix} F_x \\ F_y \\ F_z \end{pmatrix}$

$\hookrightarrow$ Compressible $\Rightarrow \nabla \cdot \vec{F} = 0$

Curl: $\nabla \times \vec{F}$

Laplacian, $\nabla^2: \nabla^2 f = \nabla \cdot \nabla f$

Surface Integrals, $\iint f(x, y, z) dS$

$dS = \left| \frac{d\vec{r}}{du} \times \frac{d\vec{r}}{dv} \right| du dv$

Surface Area = $\iint \left| \frac{d\vec{r}}{du} \times \frac{d\vec{r}}{dv} \right| du dv$

Flux = $\iint [\vec{F}(r(u, v)) \cdot \vec{N}] du dv$ Outward
 $= \iint [\vec{F}(r(u, v)) \cdot \left(\frac{d\vec{r}}{du} \times \frac{d\vec{r}}{dv} \right)] du dv$

Integrating Factor, $e^{\int p(x) dx}$

$\frac{dy}{dx} + p(x)y = Q(x)$

$\Rightarrow \frac{d}{dx} [ye^{\int p(x) dx}] = Q(x)e^{\int p(x) dx}$

Example 1:

$\cos(x) \frac{dy}{dx} - y \sin x = x^2$

$\frac{dy}{dx} + \left(\frac{-\sin x}{\cos x} \right) y = \frac{x^2}{\cos(x)}$ — ①

Let $I = e^{\int \frac{-\sin x}{\cos x} dx} = e^{\ln \cos x} = \cos x$

Multiplying both sides of ① with I ,

$\Rightarrow (\cos x) \left(\frac{dy}{dx} \right) + (-\sin x)(y) = x^2$

$\frac{d}{dx} [\cos(x)y] = x^2$

$\cos(x)y = \int x^2 dx$

$y = \frac{1}{\cos x} \left[\frac{x^3}{3} + C \right]$ *

$J(u, v, w) = \det \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{bmatrix}$

Line integral, $\int f ds$

$ds = \left| \frac{d\vec{r}}{dt} \right| dt = \sqrt{\left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2 + \left(\frac{dz}{dt} \right)^2} dt$

Scalar field: $\int_{t_1}^{t_2} [f(\vec{r}(t)) \left| \frac{d\vec{r}}{dt} \right|] dt$

Vec field / Work Done: $\int_{t_1}^{t_2} [\vec{F}(\vec{r}(t)) \cdot \frac{d\vec{r}}{dt}] dt$

$\hookrightarrow$ If $\nabla \times \vec{F} = 0$ OR $\oint \vec{F} \cdot d\vec{r} = 0$

$\hookrightarrow$ Conservative Vector Field

$\hookrightarrow \vec{F} = \nabla f$

$\hookrightarrow \int_{t_1}^{t_2} [\vec{F}(\vec{r}(t)) \cdot \frac{d\vec{r}}{dt}] dt = f(t_2) - f(t_1)$

Gauss / Divergence

$\iint_S [\vec{F} \cdot \vec{n}] dS = \iiint_V [\nabla \cdot \vec{F}] dV$

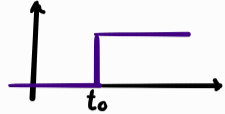
Stokes 3D

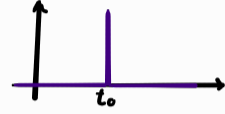
$\iint [(\nabla \times \vec{F}) \cdot \vec{n}] dS = \int \vec{F} \cdot d\vec{r}$

Stokes 2D / Green's

$\iint_A \left[\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right] dA = \int [P \frac{dx}{dt} + Q \frac{dy}{dt}] dt$

$\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty f(t) e^{-st} dt$

$u(t - t_0) = \begin{cases} 0 & \text{if } t < t_0 \\ 1 & \text{if } t > t_0 \end{cases}$ 

$\delta(t - t_0) = \begin{cases} \infty & \text{if } t = t_0 \\ 0 & \text{if } t \neq t_0 \end{cases}$ 

$\hookrightarrow \int_{t_0-\epsilon}^{t_0+\epsilon} \delta(t - t_0) dt = 1$

$f(x) = a_0 + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi}{L}x\right) + b_n \sin\left(\frac{n\pi}{L}x\right) \right]$

$L = \text{Half-Period} = \frac{1}{2}T$

$a_0 = \frac{1}{2L} \int_{-L}^L f(x) dx$

$a_0 + \sum_{n=1}^N \left[a_n \cos\left(\frac{n\pi}{L}x\right) + b_n \sin\left(\frac{n\pi}{L}x\right) \right]$

$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi}{L}x\right) dx = 0$ if $f(x) = \text{odd}$
 $\hookrightarrow f(x) = -[f(-x)]$

$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi}{L}x\right) dx = 0$ if $f(x) = \text{Even}$
 $\hookrightarrow f(x) = f(-x)$

Value of discriminant	Type of roots for the auxiliary equation	Complementary function
$b^2 - 4ac > 0$	Real, distinct roots, λ_1 and λ_2	$y = Ae^{\lambda_1 x} + Be^{\lambda_2 x}$
$b^2 - 4ac = 0$	Real, repeated root, λ	$y = (A + Bx)e^{\lambda x}$ or $y = A \cosh(\lambda x) + B \sinh(\lambda x)$
$b^2 - 4ac < 0$	Complex roots: (a) $\alpha \pm \beta i$, with $\alpha \neq 0$ (b) $\pm \beta i$ (i.e. purely imaginary roots)	(a) $y = e^{\alpha x} (P \cos \beta x + Q \sin \beta x)$ (b) $y = P \cos \beta x + Q \sin \beta x$